Genome, Environment, Microbiome and Metabolome in Autism (GEMMA) Study: Beyond Biomarker Identification for the Precision Treatment and Early Detection of Autism Spectrum Disorders

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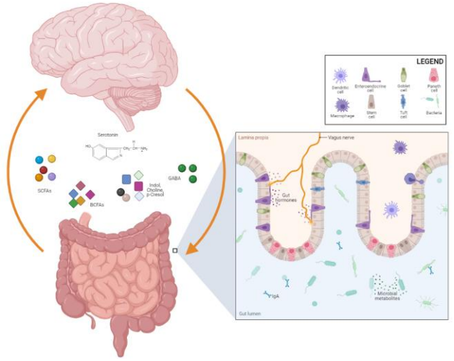
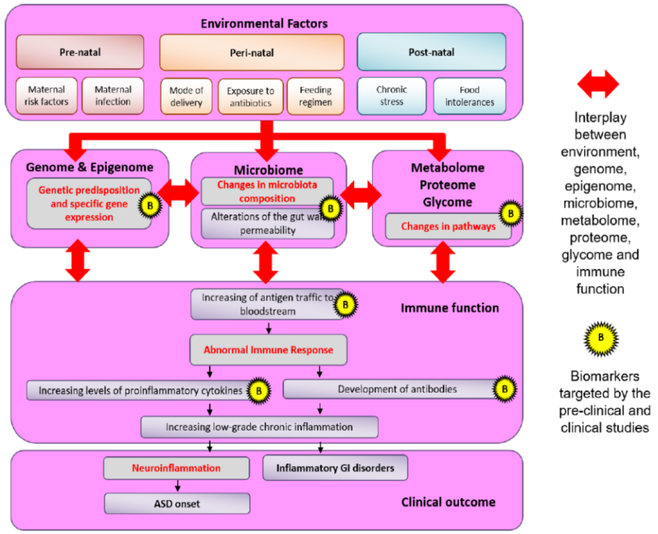
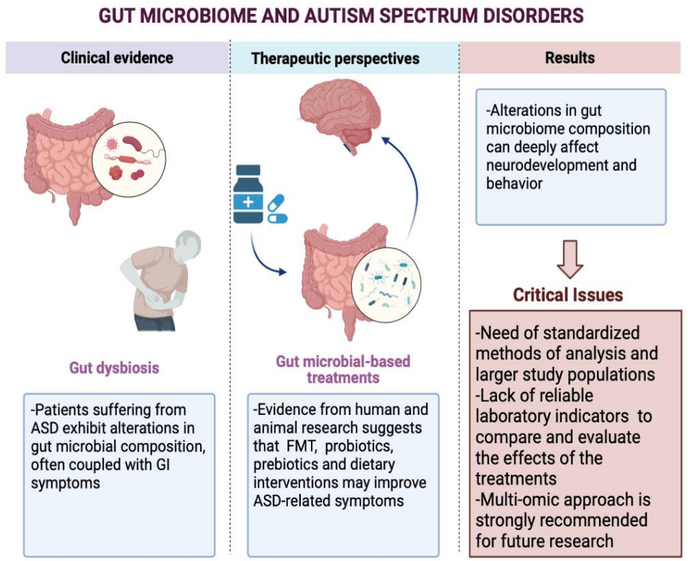
Autism Spectrum Disorder (ASD) is a term used to define a range of neurodevelopmental disorders denoted by repeated behaviours that persist into adulthood and cause difficulties in communication and social interaction. The risk of developing such a condition increases 10-fold when a child is born in a family where there is already a sibling affected by the same disease. Therefore, it becomes important to carry out studies that quickly detect the presence of biomarkers that could be signs of the possible development of ASD. It is noteworthy that gastrointestinal and immune system disorders are often associated with this condition.



This long-term study of the evolution of the microbiota and metabolomic profile will therefore help to define a rapid prognosis of ASD by identifying characteristic metabolomic biomarkers allowing for immediate and early treatment.



The GEMMA (Genome, Environment, Microbiome and Metabolome in Autism) project aims to evaluate the influence of external factors in the alteration of the intestinal microbiota and how the genome/metagenome interaction can cause an immune response due to occurrence of epigenetic and metabolomic alterations, facilitating and consequently increasing intestinal molecular traffic and contributing to the onset of ASD. GEMMA aims to carry out metabolomic analysis on blood, urine, stool and saliva samples collected from 600 subjects and coming from different parts of the world (Irish Center for Autism and neurodevelopmental Research (ICAN), Local Health Authority (ASL) in Italy and Massachusetts General Hospital for Children (MGHFC) Lurie Centre for Autism in the USA) in order to identify both the altered molecular pathways in patients with ASD and those useful for the creation of a metabolomic phenotype of the GI microbiota. Data analysis will help reveal the relationship between gut microbiota and the host.



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